

A Packing of 11 Four-Spaces in $\mathbb{R}^9$

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Abstract

We give 11 4-dimensional subspaces of $\mathbb{R}^9$ whose smallest squared chordal distance is at least 2.444131. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.423926, obtained by deleting 1 of the 12 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 11 subspaces of dimension 4 in $\mathbb{R}^9$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.444131 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 9$, $n = 4$. The cell $N = 11$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 2.423926, obtained by deleting 1 of the 12 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(9 - 4)/9 \cdot 11/10 = 2.444444$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

```
python grassmann/verify.py grassmann/packings/m9n4N11.txt
```

in <https://github.com/Ge-limin/math>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 61 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
152758235856724069934520047167227800081743720903225074753281794139290040
289353868021706456251494198531866695744038727385145336648667900547075719
55850348990059985399394204550610924404475987183
```

denominator:

```
624999999998896042946627679088042388340974226239563490421196707837717924
423103065616746602007134598196955570830612295566884295041732295393886329
1520576810672456286553881372938395204654784559
```

B Coordinates

11 blocks of 9 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m9n4N11.txt`.

```
-36788810183 519147167872 -344546461459 -4788310758263927396362 -311517749804 136394312760 29663648206413814336952 231753654409 141762390116 30257118334
-303387984400 234651056552 216030016046 -4871118023350721801485 -644378274312 -187161752446 -40591905798589358944 374353938791 249717081883 -414015619044
-200511657434 -231898833625 658508023715 -603519276338938009974 461731549586 -54768304206 57910026953 102540620375 386615411848 -366250000291 -557185037824
-372027686850 500667065397 231605582635 45227664280414252474870 207862645258 -402824829434 -64030954199136875281688 371007829383 -326134087247 14746985341
-29482854967 -420092192566 16618972738 261230349325-124181005776 -55453753708 283382392174 -162825580298373647308 -61020282669 -689818922386 252258467530
244941913034 -103811274194 215557201899 303658967392686182590 25697118886 -181351578312 -27021034642341778409353 73833208295 307677573257 257004997238
-241756075700 -328184761584 -201467206432 -7894554930668585756 -186424762193 66447949835 10784070874632726496436 389262922442 -126417797038 276086848876
771162358129 180361643385 145560132666 -1136237566306690589302 -319484846346 -260024571920 -112853315293180840390 265507678282 699967396968 291980062913
118277756094 208544310145 488699212014 141299403415-348819255599 246034765280 -404844824922 107782855609965576261 -167774975590 80640195721 -580894417141
-432087841204 212275649584 24091819426 -1701500545491568116566 -185199208394 -106873828222 4263756512063341531775 151301731065 189241834367 -591595749579
60743915455 553716924074 567655082324 169322219052 322447694101 538177069361 623294611267 -149163468582160453284249 319621285498 -130253335685 79128742184
-411318142495 516190341070 59901327762 -118257433695794666175 605303758939 -297812950930 5260193607502563062972 -536321625783 158867747839 44600619637
511588002727 50200702860 230135639562 388333091660 79573636673 342199386895 635443360879 -149789830510-508975966288 450767961442 317828728035 -331668115558
72462904485 183236011095 -166025640154 38012071517872593708147 -176949981761 139546403621 28923794894435841766977 -48597139424 -248300554212 -116185852725
-248946440541 -394536114770 686000328966 -1246885512138433658012 348487570309 160008564985 6840135110389432587158 362749644079 -496865889739 -139091249154
344491161402 -62636436446 93725537546 -668786141991-10538585091 494001176570 171222050663 -21958436072426180956250 64499926763 78239551762 218411399587
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-345945006471 -96314225742 50782310000 -608910882804467899362 137995826987 -423125082924 32071611059315536892113 3196489439 443079947344 -278290063942
-297780194911 340271883441 -306569907630 5601998892677481696508 425093761382 235773023334 -22833016607-469976926118 -591622231393 268099870308 53046902241
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231581646593 -96503238400 124494540551 207824707195-465062746898 -243739679605 106273964384 4406465464212943554721 168620589395 475950138966 483552198819
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-153765808043 -814896087253 -165256668364 19578683372837685868 -243101905535 548634641459 -1875087988037064786814 294103396152 153376909436 -352142326376
-637435699279 41786191090 -204010259535 227588952857445546046773 -108751877701 149628623360 -302976890302892523509 606325296135 -293160606822 -28076414369
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-355758853822 -79293246794 477421775796 -588093022851046372588 5218325965 -27377010457 -310463487826 -745299301473 261981390030 175137165798 -82775100015
-282988900943 -79795592408 -760107389874 -160974091936575617979 676002032958 -4686584692 538277915657 439250773657 202640699036 23977833922 -165489267680
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```